**University of Florida, Department of Mathematics**

# Analysis PhD Exam, September 2000

*Write all proofs in a neat and logical fashion.*

**1.** State and prove the Radon–Nikodym Theorem.

**2.** State and prove the Vitali Integral Convergence Theorem.

**3.** Let $(X, \Sigma, \mu)$ be a finite measure space and suppose $\Sigma = \sigma(\Sigma_0)$, where $\Sigma_0$ is an algebra. Show that for every $\varepsilon > 0$, if $E \in \Sigma$, then there exists an $E_0 \in \Sigma_0$ such that $\mu(E \triangle E_0) < \varepsilon$.

**4.** Let $Y$ and $Z$ be closed subspaces in the B-space $X$. Suppose each $x \in X$ has a unique representation in the form $x = y + z$, with $y \in Y$ and $z \in Z$. Show that there exists a constant $K$ such that $\|y\| \leq K\|x\|$ and $\|z\| \leq K\|x\|$, for each $x \in X$. [Hint: First apply the Closed Graph Theorem to $T : X \to Y$, $T(x) = y$.]

**5.** Let $(X, \Sigma, \mu)$ be a measure space. Let $(f_n)$ be a sequence of integrable functions such that $\sum_n \int |f_n| \, d\mu < \infty$. Does $\sum_n f_n$ converge a.e.? Prove.

**6.** Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X$ and $Y$ be integrable random variables. Suppose $X$ and $Y$ are independent. Find $E(X/Y)$.

**7.** Let $(X, \Sigma, \mu)$ be a finite measure space; let $(f_n)$ be a sequence of integrable functions such that $\int_E f_n \, d\mu$ converges for each $E \in \Sigma$. Assume $(f_n)$ is $L^1$-bounded. Show there exists an integrable function $f$ such that $\int_E f_n \, d\mu \to \int_E f \, d\mu$ for all $E \in \Sigma$.